

usepackage[a4paper,margin=0.65in]geometry

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165.

ON SOME GEOMETRICAL THEOREMS RELATING TO A TRIANGLE CIRCUMSCRIBED ABOUT A CONIC.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 169–175.]

THE following investigations were suggested to me by Sir F. Pollock's interesting paper "On a Geometrical Theorem relating to an Equilateral Triangle circumscribed about a Circle," [*Quart. Math. Jour.* t. I. (1857), pp. 167–169].

If on the sides of a triangle ABC , there be taken points α, β, γ , such that $A\alpha, B\beta, C\gamma$ meet in a point O ; and if on each side of the triangle there be taken two points forming with the two angles on the same side an involution having the first-mentioned point on the same side for a double point; then if three of the six points lie in a line, the two lines are said to be harmonically related with respect to the triangle ABC and point O . Call these the lines $(r), (s)$.

The triangle ABC and point O give rise to a determinate conic; viz. the conic touching the sides at the points α, β, γ .

The harmonic of the point O with respect to the triangle is a line (o) , which is also the polar of O with respect to the conic. The conic meets this line in two points (as the figure is drawn, imaginary ones), I, J .

Suppose now that the harmonic lines $(r), (s)$ are harmonically related to the points I, J , (i.e. let the lines $(r), (s)$ and the lines through their point of intersection and I, J , be a harmonic pencil, (or, what is the same thing, let I, J , and the points in which the line of junction meets the lines $(r), (s)$ be a harmonic range), then,

Theorem, the point of intersection of the lines $(r), (s)$ lies on the conic.

In order to prove this, take

$$x = 0, \quad y = 0, \quad z = 0, \quad \text{for the equations of } BC, CA, AB,$$

$$x = y = z, \quad \text{for the point } O,$$

the equations of the harmonic lines will be

$$ax + by + cz = 0,$$

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 0,$$

the equation of the conic is

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0;$$

the equation of the line (o) is

$$x + y + z = 0.$$

The coordinates of the points I, J , are

$$x : y : z = 1 : \omega : \omega^2,$$

and

$$x : y : z = 1 : \omega^2 : \omega,$$

where ω is an imaginary cube root of unity.

The equations of the lines joining the point of intersection of $(r), (s)$ with the points I, J , are

$$\frac{ax + by + cz}{a + b\omega + c\omega^2} = \frac{bcx + cay + abz}{bc + caw + ab\omega^2}$$

and

$$\frac{ax + by + cz}{a + b\omega^2 + c\omega} = \frac{bcx + cay + abz}{bc + ca\omega^2 + ab\omega},$$

and these will form with the lines

$$ax + by + cz = 0,$$

$$bcx + cay + abz = 0,$$

a harmonic pencil, if

$$(a + b\omega + c\omega^2)(bc + caw^2 + ab\omega) + (a + b\omega^2 + c\omega)(bc + caw + ab\omega^2) = 0,$$

i.e. if

$$6abc - (bc^2 + b^2c + ca^2 + c^2a + ab^2 + a^2b) = 0;$$

which is, therefore, the condition in order that (r) , (s) , I , J may be harmonically related.

The coordinates of the point of intersection of (r) , (s) are

$$x : y : z = a(b^2 - c^2) : b(c^2 - a^2) : c(a^2 - b^2);$$

and substituting these values in

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy,$$

we see, *à priori*, that the result contains the factor

$$2abc + bc^2 + b^2c + ca^2 + c^2a + ab^2 + a^2b = (b + c)(c + a)(a + b);$$

in fact $b + c = 0$ gives $x : y : z = 0 : b(b^2 - a^2) : b(a^2 - b^2)$, i.e. $x = 0$, $y - z = 0$, which makes $x^2 + y^2 + z^2 - 2yz - 2zx - 2xy$ vanish; and so for the other factors.

Effecting the substitution, we find

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = (2abc + bc^2 + b^2c + ca^2 + c^2a + ab^2 + a^2b) \times (-6abc + bc^2 + b^2c + ca^2 + c^2a + ab^2 + a^2b)$$

which equals 0, on account of the second factor.

Hence the point of intersection of the lines (r) , (s) lies upon the conic. Q.E.D.

Consider, next, a side of the triangle: then, *Theorem*, the following four points on a side of the triangle, viz. the point of contact, the point of intersection with the line (o) , and the points of intersection with the harmonic lines (r) , (s) , form a harmonic range.

In fact, for the side $x = 0$ the points in question are given by means of the equations

$$y - z = 0, \quad y + z = 0, \quad by + cz = 0, \quad \frac{y}{b} + \frac{z}{c} = 0,$$

and forming the equations

$$y^2 - z^2 = 0, \\ y^2 + \left(\frac{b}{c} + \frac{c}{b}\right)yz + z^2 = 0,$$

the theorem is at once seen to be true. Q.E.D.

It is interesting to give the analytical expressions for some other of the points and lines of the figure. The harmonic lines (r) , (s) , meet in a point of the conic given by the equations

$$\xi : \eta : \zeta = a(b^2 - c^2) : b(c^2 - a^2) : c(a^2 - b^2).$$

This point may be spoken of as the common point of intersection of the harmonic lines with the conic. The foregoing values give

$$\xi'' = \eta - \zeta'' = a\xi \left(\frac{b}{c} + \frac{c}{b} \right),$$

$$\eta'' - \zeta'' - \xi'' = b\eta \left(\frac{c}{a} + \frac{a}{c} \right),$$

$$\zeta'' - \xi'' - \eta'' = c\zeta \left(\frac{a}{b} + \frac{b}{a} \right),$$

which may also be written

$$\xi'' = (\eta - \zeta)^2 = a\xi \frac{1}{bc}(b + c)^2 = QR\xi,$$

$$\eta'' = (\zeta - \xi)^2 = b\eta \frac{1}{ca}(c + a)^2 = RP\eta,$$

$$\zeta'' = (\xi - \eta)^2 = c\zeta \frac{1}{ab}(a + b)^2 = PQ\zeta,$$

if for shortness

$$-\xi + \eta + \zeta = P,$$

$$\xi - \eta + \zeta = Q,$$

$$\xi + \eta - \zeta = R,$$

formulae which will be presently useful.

Each harmonic line meets the conic in another point, which may be termed the simple point of intersection. For the line $ax + by + cz = 0$, the coordinates of the simple point of intersection are given by $x : y : z = \frac{1}{a}\xi(b + c) : \frac{1}{b}\eta(c + a) : \frac{1}{c}\zeta(a + b)$; or, what is the same thing, by

$$x : y : z = \frac{1}{Pa} : \frac{1}{Qb} : \frac{1}{Rc},$$

and, in like manner, the coordinates of the simple point of intersection of the line

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 0$$

are given by

$$x : y : z = \frac{a}{P} : \frac{b}{Q} : \frac{c}{R}.$$

Hence the equation of the line joining the two simple points of intersection is

$$\begin{vmatrix} x, & y, & z \\ \frac{1}{Pa}, & \frac{1}{Qb}, & \frac{1}{Rc} \\ \frac{a}{P}, & \frac{b}{Q}, & \frac{c}{R} \end{vmatrix} = 0,$$

or expanding and reducing

$$Px + Qy + Rz = 0.$$

The equation of the tangent to the conic at the common point of intersection is evidently

$$Px + Qy + Rz = 0.$$

The last-mentioned lines, together with the harmonic lines (r) , (s) , viz. the lines

$$\begin{aligned} ax + by + cz &= 0, \\ \frac{x}{a} + \frac{y}{b} + \frac{z}{c} &= 0, \end{aligned}$$

may be considered as the sides of an inscribed quadrilateral; the equation of the conic must therefore be expressible in a form in which this is put in evidence: to do this, I first form the equation

$$(ax + by + cz) \left(\frac{x}{a} + \frac{y}{b} + \frac{z}{c} \right) = x^2 + y^2 + z^2 + 2 \left(1 - \frac{QR}{bc} \right) yz + 2 \left(1 - \frac{RP}{ca} \right) zx + 2 \left(1 - \frac{PQ}{ab} \right) xy,$$

which may also be written

$$(ax + by + cz) \left(\frac{x}{a} + \frac{y}{b} + \frac{z}{c} \right) = x^2 + y^2 + z^2 - 2yz - 2zx - 2xy,$$

where

$$\lambda = \frac{a^2 + b^2 - c^2}{2ab}, \quad \mu = \frac{b^2 + c^2 - a^2}{2bc}, \quad \nu = \frac{c^2 + a^2 - b^2}{2ca},$$

and then putting

$$\Delta = P^2 + Q^2 + R^2 - 2QR - 2RP - 2PQ,$$

an equation which gives

$$\begin{aligned} \Delta &= P^2 - 4QR\xi, \\ &= Q^2 - 4RP\eta, \\ &= R^2 - 4PQ\zeta, \end{aligned}$$

and

$$-PQR - 8\xi\eta\zeta = (\xi + \eta + \zeta) \Delta,$$

it is easy by means of these relations to verify the identical equation

$$M(ax+by+cz)\left(\frac{x}{a}+\frac{y}{b}+\frac{z}{c}\right)+(P\xi x+Q\eta y+R\zeta z)(Px+Qy+Rz) = (P^2+Q^2+R^2+2QR+2RP+2PQ)(x^2+y^2+z^2+2xy+2yz+2zx),$$

or, writing for Δ its value, $\Delta = 0$, the equation of the conic takes the form

$$M(ax+by+cz)\left(\frac{x}{a}+\frac{y}{b}+\frac{z}{c}\right) - (P\xi x+Q\eta y+R\zeta z)(Px+Qy+Rz) = 0,$$

which is as it should be.

It may be added, that the common point of intersection and the points in which the harmonic lines (r), (s) meet a side of the triangle lie in a conic passing through the points I , J , such that with respect to this conic the point of contact is the pole of the line (o). Thus for the side $x = 0$ the equation of the conic in question is

$$\theta x(y+z) + (1 - \frac{1}{2}\theta) x^2 + (y+z)\left(\frac{y}{b} + \frac{z}{c}\right) = 0,$$

where

$$\theta = \frac{a(b-c)^2(b+c) + 2bc}{bc(a-b)(a-c)} \cdot \frac{(b+c)^2}{2bc}.$$

and similarly for the other two conics.

In the case where the triangle is an equilateral triangle and the line (o) is the line at ∞ , the conic becomes a circle, the points I , J , are the circular points at infinity, and lines harmonic in respect to the points I , J , become lines at right angles to each other: the foregoing results agree, therefore, with Sir F. Pollock's Theorem.

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NOTE ON THE HOMOLOGY OF SETS.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), p. 178.]

LET L denote a set of any four elements a, b, c, d , and in like manner Λ, L_1 , &c. sets of the four elements $\alpha, \beta, \gamma, \delta; a_1, b_1, c_1, d_1$, &c.; then we may establish a relation of homology between four sets L, L_1, L_2, L_3 , and four other sets $\Lambda, \Lambda_1, \Lambda_2, \Lambda_3$; viz. considering the corresponding anharmonic ratios of the different sets, we may suppose a relation of homology between these ratios. Thus considering the set L , to write

$$\begin{aligned} x &= (a - b)(c - d), \\ y &= (a - c)(d - b), \\ z &= (a - d)(b - c), \end{aligned}$$

then $x + y + z = 0$ and the anharmonic ratios of the set are $x : y : z$; we may, if we please, take $x : y$ as the anharmonic ratio of the set. And in like manner taking $\xi : \eta$ as the anharmonic ratio of the set $\alpha, \beta, \gamma, \delta$, &c., the assumed relation between the sets L, L_1, L_2, L_3 and the sets $\Lambda, \Lambda_1, \Lambda_2, \Lambda_3$ will be

$$\begin{vmatrix} x\xi, & x\eta, & y\xi, & y\eta \\ x_1\xi_1, & x_1\eta_1, & y_1\xi_1, & y_1\eta_1 \\ x_2\xi_2, & x_2\eta_2, & y_2\xi_2, & y_2\eta_2 \\ x_3\xi_3, & x_3\eta_3, & y_3\xi_3, & y_3\eta_3 \end{vmatrix} = 0;$$

and it is to be observed, that this relation is independent of the particular ratio $x : y$ which has been chosen as the anharmonic ratio of the set; in fact, if we write $x = -y - z$, $x_1 = -y_1 - z_1$, &c., then reducing the result by means of an elementary property of determinants, the equation will preserve its original form, but will contain the ratios $y : z; \eta : \zeta$, &c., instead of the ratios $x : y; \xi : \eta$, &c.

167.

APROPOS OF PARTITIONS.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 183–184.]

LET $\Pi(1-x^a)(1-x^b)\dots(1-x^k)$ (κ factors) and assume that $\left[\frac{1}{\Pi(1-x^a)}\right]$ is the part of $\frac{1}{\Pi(1-x^a)}$ which involves negative powers of $1-x$, then

Coefficient x^n in $\left[\frac{1}{\Pi(1-x^a)}\right] = \text{coefficient } x^{n+1} \text{ in } \left(\frac{1}{(1-x)^{n+1}}\Pi\frac{1-x}{1-x^a}\right)$,

which suggests the question of the expansion in powers of z of the function

$$\frac{1}{(1-z)^a} \Pi \frac{1-z}{1-(1-z)^a}.$$

Now by the definition of Bernoulli's numbers

$$\frac{1}{e^t - 1} = \frac{1}{t} - \frac{1}{2} + B_1 \frac{t}{1.2.3} - B_2 \frac{t^3}{1.2.3.4.5.6} + B_3 \frac{t^5}{\dots} - \&c.,$$

from which it is easy to deduce

$$\frac{t}{1 - e^{-t}} = 1 + \frac{1}{2}t + \frac{1}{12}t^2 - \frac{1}{720}t^4 + \&c.,$$

and, writing in this formula $t = a \log(1-z)$, we have

$$\frac{-a \log(1-z)}{1 - (1-z)^a} = e^{\frac{1}{2}a \log(1-z) + \frac{1}{12}a^2 \log^2(1-z) - \frac{1}{720}a^4 \log^4(1-z) + \&c.},$$

i.e.

$$\frac{z}{1 - (1 - z)^a} = \frac{1}{a} \frac{-z}{\log(1 - z)} \cdot e^{-\frac{1}{2}a \log(1-z) + \frac{1}{12}a^2 \log^2(1-z) - \frac{1}{720}a^4 \log^4(1-z) + \&c.},$$

and putting p_κ for $abc \dots$ and $S_1, S_2 \dots$ for the sums of the powers, we have, taking the product

$$\prod \frac{z}{1 - (1 - z)^a} = \frac{1}{p_\kappa} e^{\kappa \log \frac{-z}{\log(1-z)} - \frac{1}{2}S_1 \log(1-z) + \frac{1}{12}S_2 \log^2(1-z) - \frac{1}{720}S_4 \log^4(1-z) + \&c.},$$

whence also

$$\frac{1}{(1 - z)^q} \prod \frac{z}{1 - (1 - z)^a} = \frac{1}{p_\kappa} e^{\kappa \log \frac{-z}{\log(1-z)} - (q + \frac{1}{2}S_1) \log(1-z) + \frac{1}{12}S_2 \log^2(1-z) - \frac{1}{720}S_4 \log^4(1-z) + \&c.},$$

from which the development may be found.

The index of e is

$$\begin{aligned} & (q + \frac{1}{2} - \frac{1}{2}S_1)z \\ & + (\frac{1}{2}q + \frac{1}{4} - \frac{1}{4}S_1 + \frac{1}{6}S_2 - \frac{1}{12}\kappa)z^2 \\ & + (\frac{1}{3}q + \frac{1}{6} - \frac{1}{6}S_1 + \frac{1}{6}S_2 - \frac{1}{12}\kappa)z^3 \\ & + \&c. \end{aligned}$$

and developing the exponential,

1°. The coefficient of z is

$$q + \frac{1}{2}S_1 - \frac{1}{2}(\kappa - 2).$$

2°. The coefficient of z^2 is

$$\frac{1}{2}[q^2 + q\{\frac{1}{2}S_1 - \frac{1}{2}(\kappa - 3)\} + \frac{1}{4}S_1^2 - \frac{1}{6}S_2 - \frac{1}{2}(\kappa - 3)S_1 + \frac{1}{12}(\kappa - 3)(\kappa - 4)]$$

and so on.

The peculiarity is the appearance of the factors $\kappa - 2, \kappa - 3, \&c.$ If we neglect these terms, and consider as well q as $a, b, c \dots$ to be each of them of the dimension unity, the coefficients will be homogeneous.

168.

A DEMONSTRATION OF THE FUNDAMENTAL PROPERTY OF
GEODESIC LINES.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 185–6.]

THIS is the translation of a paragraph in M. Bertrand's Memoir "Démonstration géométrique de quelques théorèmes relatifs à la théorie des Surfaces," Liouv. t. XIII. (1848), p. 73, in which he shows that the theorem that the geodesic lines on a surface have at each point their osculating plane normal to the surface is an immediate consequence of Meunier's theorem.

169.

EISENSTEIN'S GEOMETRICAL PROOF OF THE FUNDAMENTAL THEOREM FOR QUADRATIC RESIDUES. (TRANSLATED FROM THE ORIGINAL MEMOIR, CRELLE, T. XXVIII. (1844), WITH AN ADDITION, BY A. CAYLEY.)

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 186–191.]

LET p be a positive odd prime, a the aggregate of all the even numbers $< p$ and > 0 , viz. $a = 2, 4, 6 \dots p-1$; and let q be any integer not divisible by the modulus p ; then if r denote the general term of the residues of the multiples qa in respect to the modulus p , it is clear that the numbers of the series the general term of which is $(-)^r r$ will coincide to multiples of p près with the numbers of the series $(-)^a a$; so that we shall have the two congruences

$$q^{\frac{1}{2}(p-1)} \Pi a \equiv \Pi r \pmod{p}, \quad \text{and} \quad \Pi a \equiv (-)^{\Sigma r} \Pi r \pmod{p},$$

from which it follows that

$$q^{\frac{1}{2}(p-1)} \equiv (-)^{\Sigma r} \pmod{p}, \quad \text{and therefore} \quad \left(\frac{q}{p}\right) = (-)^{\Sigma r}.$$

Let $E\left\{\frac{x}{y}\right\}$ denote the greatest whole number contained in the fraction $\frac{x}{y}$, then it is clear that $\Sigma qa = p \Sigma E\left(\frac{qa}{p}\right) + \Sigma r$; and since all the a 's are even, and $p \equiv 1 \pmod{2}$ it follows that $\Sigma r \equiv \Sigma E\left(\frac{qa}{p}\right) \pmod{2}$; and we have, therefore,

$$\left(\frac{q}{p}\right) = (-)^{\Sigma E(qa/p)}.$$

When $q = 2$, the formula gives at once the value of $\left(\frac{2}{p}\right)$; when, on the other hand, q is odd, and therefore $q - 1$ even, we find, by a simple transformation,

$$\Sigma E\left(\frac{qa}{p}\right) = \Sigma E\left(\frac{a}{p}\right) + \Sigma E\left(\frac{2a}{p}\right) + \Sigma E\left(\frac{(q-1)a}{p}\right) - \frac{1}{2} E\left(\frac{\frac{1}{2}(q-1)a}{p}\right) \equiv \Sigma E\left(\frac{\frac{1}{2}(q-1)a}{p}\right) \pmod{2};$$

and if the last sum be represented by μ , then also

$$\left(\frac{q}{p}\right) = (-1)^\mu.$$

Imagine now in a plane, a rectangular system of coordinates (x, y) and the whole plane divided by lines parallel to the axes at distances = 1 from each other into squares of the dimension = 1. And let the angles which do not lie on the axes of coordinates be called "lattice points."

Take, now, upon any vertical parallel a point corresponding to the ordinate y ; then $E(y)$ will denote the number of lattice points which lie between this point and the horizontal axis; and, in like manner, taking upon any horizontal parallel a point corresponding to the abscissa x , then $E(x)$ will denote the number of lattice points which lie between this point and the vertical axis. If, therefore, we draw in the plane a curve the equation of which is $y = \varphi x$, then the sum

$$E\varphi 1 + E\varphi 2 + E\varphi 3 + \&c.$$

will denote the number of lattice points which lie between the curve and the axis of x , including any below points which lie upon the curve.

Suppose now, to return to the subject, that AB represents the straight line which has for its equation $y = \frac{q}{p}x$, where p and q are now assumed to be both of them positive

odd primes; and let $AD = FB = \frac{1}{2}p$, $AF = DB = \frac{1}{2}q$, $AC = EG = \frac{1}{2}(p-1)$, $AE = CG = \frac{1}{2}(q-1)$.

Then if μ denote the number of the lattice points between AB and AD as far as the ordinate CG inclusively (these lattice points are distinguished in the figure by the mark $[*]$), by what precedes $\left(\frac{q}{p}\right) = (-)^\mu$. But since the equation of the straight line AB may also be written $x = \frac{p}{q}y$, we have in the same way, if ν denote the number of the lattice points which lie between AB and AF as far as the abscissa EG inclusively (these are distinguished in the figure by the mark $[o]$), $\left(\frac{p}{q}\right) = (-)^\nu$. But the lattice points marked with $[*]$ and $[o]$ taken together, i.e. all the lattice points to the right, and all the lattice points to the left of AB , make up the entire system of lattice points of the rectangle $AEGC$, the number of which is $\frac{1}{2}(p-1) \cdot \frac{1}{2}(q-1)$; consequently, $\mu + \nu = \frac{1}{2}(p-1) \cdot \frac{1}{2}(q-1)$, and therefore

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-)^{\mu+\nu} = (-)^{\frac{p-1}{2} \cdot \frac{q-1}{2}},$$

which is the theorem in question.

It maybe noticed that the foregoing transformation $\Sigma E\left(\frac{qa}{p}\right) \equiv \mu \pmod{.2}$ may itself be proved by the simple geometrical consideration that $\Sigma E\left(\frac{qa}{p}\right)$ is nothing else than the number of lattice points which lie on the even ordinates (those corresponding to $x = 2, 4, 6 \dots p-1$) between AB and AD as far as BD , and that each ordinate, between the axes AD and BF exclusively, contains $(q-1)$, i.e. an even number of lattice points; and besides, that the two triangles BAD and ABF are congruent, and that the latter of them stands in the same relation to BF and BD as the former does to AD and AF ; the completion of which is left to the reader.

Remark. There are figures for which simple formulæ may be obtained for the number of the interior lattice points. Imagine, for example, a circle, the centre of which lies in the axis of coordinates, and the radius of which is $\sqrt{m}$, then the number S of the lattice points which lie within the circle, including those which lie on the axes, is given by the following formula,

$$S = 1 + 4\{E(m) - E(\frac{1}{3}m) + E(\frac{1}{5}m) - \&c.\}$$

continued until the series stops of itself. It is easy to see that the equation expresses a relation between the number of lattice points of the circle and the number of lattice points of a segment included between two rectangular hyperbolas. Writing in the formula

$$\frac{S}{m} = \frac{1}{m} + 4 \left\{ \frac{Em}{m} - \frac{E\frac{1}{3}m}{m} + \frac{E\frac{1}{5}m}{m} - \&c. \right\}$$

$m = \infty$, the left-hand side becomes equal to π , and the right-hand side becomes equal to $4(1 - \frac{1}{3} + \frac{1}{5} - \&c.)$, which gives the formula of Leibnitz. There are similar

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formulæ for the number of lattice points of a system of sectors of ellipses or hyperbolas; and similar relations exist also in space, and in cases of more than three dimensions. We shall return to this important subject, which has the closest connection with the properties of the higher forms, upon another occasion.

(*Addition by the Translator.*) Eisenstein is now, alas! dead; too soon for the complete development of his various and profound researches in elliptic functions and the theory of numbers; and the promise at the conclusion of the foregoing memoir has not, I believe, been fulfilled. The formula in the Remark must, I think, have been established by geometrical considerations, and would have served to give the number of decompositions of a number into the sum of two squares; but, as I do not perceive how this is to be done, I shall follow a reverse course, and establish the theorem from considerations founded on the theory of numbers. I remark, first, that the number of lattice points in a quadrant of the circle, inclusive of those on the vertical axis, but exclusive of those on the horizontal axis and of the centre, is equal to $E\sqrt{m} + E\sqrt{(m-1)} + E\sqrt{(m-4)} + E\sqrt{(m-9)} + \&c.$; and that this sum, multiplied by four and increased by unity, gives precisely the number of lattice points of the circle, including those on the horizontal and vertical axes. The formula to be established is therefore

$$E\sqrt{m} + E\sqrt{(m-1)} + E\sqrt{(m-4)} + E\sqrt{(m-9)} + \&c. = E(m) - E(\tfrac{1}{3}m) + E(\tfrac{1}{5}m) - E(\tfrac{1}{7}m) + \&c.$$

where, as already noticed, the left-hand side denotes the number of lattice points of a quadrant of the circle, inclusive of those on the vertical axis, but exclusive of those on the horizontal axis and of the centre.

Let X_n be the number of ways in which the integer number n can be expressed as the sum of two squares. {If $n = \alpha^2 + \beta^2$, then if α and β are unequal, and neither of them is zero, this counts as two decompositions, viz. $x = \alpha, y = \beta$ or $x = \beta, y = \alpha$; but if $\alpha = \beta$, this counts only as a single decomposition; or if either of the numbers α, β , e.g. α , is zero, then, since $y = -\beta$ is excluded, this counts as a single decomposition, $x = 0, y = \beta$.} X_n will denote the number of lattice points on the quadrant of the circle radius $\sqrt{n}$. Suppose also that $E\left\{\frac{1}{k}n\right\}$ stands for unity or zero, according as $\frac{1}{k}n$ is or is not an integer. Then as m passes through the integer number n , i.e. from a value between $n-1$ and n to a value between n and $n+1$, the left-hand side of the equation is increased by X_n , and the right-hand side of

the equation is increased by $E(n) - E(\frac{1}{3}n) + E(\frac{1}{5}n) - \&c.$ We ought therefore to have

$$X_n = E(n) - E(\frac{1}{3}n) + E(\frac{1}{5}n) - E(\frac{1}{7}n) + \&c.,$$

and conversely from this equation, the original equation will at once follow. The right-hand side denotes, it should be observed, the number of factors of n of the form $\equiv 1 \pmod{4}$, less the number of factors of the form $\equiv 3 \pmod{4}$. Let $n = 2^\lambda f^\alpha f'^{\alpha'} \dots g^\beta g'^{\beta'} \dots$, where $f, f' \dots$ are odd primes $\equiv 1 \pmod{4}$ and $g, g' \dots$ are odd primes $\equiv 3 \pmod{4}$. Consider first the factors $g, g' \dots$, and forming the product

$$(1 + g + g^2 + \dots + g^\beta)(1 + g' + g'^2 + \dots + g'^{\beta'}) \dots$$

it is easy to see that if all or any one or more of the indices $\beta, \beta' \dots$ are odd, then the number of terms of the product which are $\equiv 1 \pmod{4}$ is equal to the number of terms of the product which are $\equiv 3 \pmod{4}$; but if all the indices $\beta, \beta' \dots$ are even, then the number of terms of the first form is greater by unity than the number of terms of the second form. Now the terms of the product $(1 + f + \dots + f^\alpha)(1 + f' + \dots + f'^{\alpha'}) \dots$ are all $\equiv 1 \pmod{4}$, and the number of terms is $(1 + \alpha)(1 + \alpha') \dots$. Hence if all or any one or more of the indices $\beta, \beta' \dots$ are odd, then the number of factors of n of the first form less the number of factors of the second form is zero; but if the indices $\beta, \beta' \dots$ are all even, the number of factors of the first form less the number of factors of the second form is $(1 + \alpha)(1 + \alpha') \dots$, i.e. we have

$$E(n) - E(\tfrac{1}{3}n) + E(\tfrac{1}{5}n) - \&c. = 0, \quad \text{or} \quad = (1 + \alpha)(1 + \alpha') \dots,$$

according as $\beta, \beta' \dots$ are all or any one or more of them odd, or according as they are all of them even. Now it is well known that the number $n = 2^\lambda f^\alpha f'^{\alpha'} \dots g^\beta g'^{\beta'} \dots$ does not, in the case of all or any one or more of the indices $\beta, \beta' \dots$ being odd, admit of decomposition into two squares, i.e. in this case $X_n = 0$; but if the indices $\beta, \beta' \dots$ are all even, then the number n will admit of precisely as many decompositions into two squares as the number $n' = 2^\lambda f^\alpha f'^{\alpha'} \dots$ (in fact, the only decompositions of n are those obtained from the decompositions of n' by multiplying the roots into the common factor $g^{\beta/2} g'^{\beta'/2} \dots$), and the number of decompositions of n' is moreover equal to the number of decompositions of $n'' = f^\alpha f'^{\alpha'} \dots$, which last number is in fact the product of the numbers of decompositions of $f^\alpha, f'^{\alpha'} \dots$; the number of decompositions of n into two squares (estimated according to the foregoing convention) is thus shown to be, in the case of $\beta, \beta' \dots$, all of them even, equal to $(1 + \alpha)(1 + \alpha') \dots$. And we have therefore in every case

$$X_n = E(n) - E(\tfrac{1}{3}n) + E(\tfrac{1}{5}n) - \&c.,$$

and the principal theorem is thus shown to be true.

170.

ON SCHELLBACH'S SOLUTION OF MALFATTI'S PROBLEM.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 222–226.]

THE following elegant solution of Malfatti's Problem as applied to spherical triangles is given by Dr Schellbach (*Crelle*, t. XLV. (1853), p. 186); for the reason which will be mentioned I have made a change of notation.

In a spherical triangle ABC to describe three small circles, each of them touching the other two, and also two sides of the triangle.

Let the sides of the triangle be a, b, c , and let x, y, z , be the distances of the points of contact from the adjacent angles of the triangle. Then writing

$$a + b + c = 2s,$$

$$a - x = \xi, \quad b - y = \eta, \quad c - z = \zeta,$$

whence

$$a + b + c = 2s,$$

and putting also $a = \frac{1}{2}s - l$, $b = \frac{1}{2}s - m$, $c = \frac{1}{2}s - n$,

$$l + m + n = \frac{1}{2}s,$$

it is easy to obtain

$$\frac{\cos l \cos \xi \cos \eta}{\cos \frac{1}{2}s} = \frac{\sin l \sin \xi \sin \eta}{\sin \frac{1}{2}s} = 1,$$

$$\frac{\cos m \cos \eta \cos \zeta}{\cos \frac{1}{2}s} = \frac{\sin m \sin \eta \sin \zeta}{\sin \frac{1}{2}s} = 1,$$

$$\frac{\cos n \cos \zeta \cos \xi}{\cos \frac{1}{2}s} = \frac{\sin n \sin \zeta \sin \xi}{\sin \frac{1}{2}s} = 1,$$

from which equations the unknown quantities ξ, η, ζ are to be determined. And the equations may be solved without assuming the existence of the relation $l + m + n = \frac{1}{2}s$.

To solve the equations, let the subsidiary angles λ, μ, ν , be determined by the conditions

$$\begin{aligned}\frac{\cos \lambda \cos m \cos n}{\cos \frac{1}{2}s} &= \frac{\sin \lambda \sin m \sin n}{\sin \frac{1}{2}s} = 1, \\ \frac{\cos \mu \cos n \cos l}{\cos \frac{1}{2}s} &= \frac{\sin \mu \sin n \sin l}{\sin \frac{1}{2}s} = 1, \\ \frac{\cos \nu \cos l \cos m}{\cos \frac{1}{2}s} &= \frac{\sin \nu \sin l \sin m}{\sin \frac{1}{2}s} = 1,\end{aligned}$$

then it may be shown that

$$\begin{aligned}\cos(\eta + \zeta) &= \frac{\cos \frac{1}{2}(\lambda + \eta - \zeta)}{\cos \frac{1}{2}(\lambda + l)}, & \cos(\zeta - \eta) &= \frac{\cos \frac{1}{2}(\lambda - \eta + \zeta)}{\cos \frac{1}{2}(\lambda + l)}, \\ \cos(\zeta + \xi) &= \frac{\cos \frac{1}{2}(\mu + \zeta - \xi)}{\cos \frac{1}{2}(\mu + m)}, & \cos(\xi - \zeta) &= \frac{\cos \frac{1}{2}(\mu - \zeta + \xi)}{\cos \frac{1}{2}(\mu + m)}, \\ \cos(\xi + \eta) &= \frac{\cos \frac{1}{2}(\nu + \xi - \eta)}{\cos \frac{1}{2}(\nu + n)}, & \cos(\xi - \eta) &= \frac{\cos \frac{1}{2}(\nu - \xi + \eta)}{\cos \frac{1}{2}(\nu + n)}.\end{aligned}$$

If we write

$$\begin{aligned}\tan \frac{1}{2}\phi &= \tan u \tan v \cot l, \\ \tan \frac{1}{2}\chi &= \tan u \tan v \cot m, \\ \tan \frac{1}{2}\psi &= \tan u \tan v \cot n,\end{aligned}$$

then

$$\begin{aligned}\cos(\lambda - \phi) &= \frac{\cos \frac{1}{2}\phi}{\cos u \cos v} \lambda, \\ \cos(\mu - \chi) &= \frac{\cos \frac{1}{2}\chi}{\cos u \cos v} \chi, \\ \cos(\nu - \psi) &= \frac{\cos \frac{1}{2}\psi}{\cos u \cos v} \psi,\end{aligned}$$

equations which give the values of λ, μ, ν , from which ξ, η, ζ are determined as above.

If we suppose that the sides become indefinitely small, we have the case of a plane triangle, and the equations then are

$$\eta^2 + \zeta^2 + \frac{2l}{s} \eta\zeta = l^2 - l\xi^2,$$

$$\zeta^2 + \xi^2 + \frac{2m}{s} \zeta\xi = m^2 - m\eta^2,$$

$$\xi^2 + \eta^2 + \frac{2n}{s} \xi\eta = n^2 - n\zeta^2.$$

We have here

$$(\eta + \zeta)^2 = \frac{1}{s}(s-a)yz = \left(\frac{1}{2}s-l\right)^2 - l\left\{\left(\frac{1}{2}s-y\right) + \left(\frac{1}{2}s-z\right)\right\}^2 - \frac{1}{s}\left\{\left(\frac{1}{2}s-y\right) - \left(\frac{1}{2}s-z\right)\right\}^2$$

$$(\eta - \zeta)^2 = \left(\frac{1}{2}s - y - \frac{1}{2}s + z\right)^2 = (y - z)^2,$$

and consequently

$$\eta^2 + \zeta^2 = \frac{1}{2} l^2 - 2lxl, \quad \eta\zeta = (l-l)(l-l),$$

where if

$$\phi = \frac{2mn}{s}, \quad (\lambda - \phi)l = \frac{1}{2}l + \frac{1}{2}\sqrt{s^2 - 4m^2}\sqrt{s^2 - 4n^2} - 4mn,$$

i.e.

$$\begin{aligned} \lambda &= \frac{2mn}{s} + \frac{1}{s} \sqrt{l^2 + \frac{4m^2n^2}{l^2} - m^2 - n^2} \\ &= \frac{2mn}{s} + \frac{1}{2sl} \sqrt{(s^2 - 4m^2)(s^2 - 4n^2)}. \end{aligned}$$

Hence

$$\begin{aligned} \eta^2 + \zeta^2 &= 2nl - \frac{4lmn}{s} - sl + \frac{1}{2}\sqrt{s^2 - 4m^2}\sqrt{s^2 - 4n^2}, \\ \eta\zeta &= 2nl - sn + \frac{1}{2}\sqrt{s^2 - 4l^2}\sqrt{s^2 - 4n^2}, \\ \zeta^2 + \xi^2 &= 2ml - \frac{4lmn}{s} - sm + \frac{1}{2}\sqrt{s^2 - 4n^2}\sqrt{s^2 - 4l^2}, \\ \zeta\xi &= 2ml - sl + \frac{1}{2}\sqrt{s^2 - 4m^2}\sqrt{s^2 - 4n^2}, \\ \xi^2 + \eta^2 &= 2ml - \frac{4lmn}{s} - sn + \frac{1}{2}\sqrt{s^2 - 4l^2}\sqrt{s^2 - 4m^2}, \\ \xi\eta &= 2ml - sm + \frac{1}{2}\sqrt{s^2 - 4l^2}\sqrt{s^2 - 4m^2}, \end{aligned}$$

which is in fact at once deducible from the formulæ in my paper “On a System of Equations connected with Malfatti's Problem and on another Algebraical System,” (*Camb. and Dubl. Math. Journ.* t. IV. (1849), p. 270 [79]).

Write now for $l, m, n, \xi, \eta, \zeta$, their values in terms of a, b, c, x, y, z . We have

$$\left(\frac{1}{2}s - y\right)^2 + \left(\frac{1}{2}s - z\right)^2 + \frac{2}{s}\left(\frac{1}{2}s - a\right)\left(\frac{1}{2}s - y\right)\left(\frac{1}{2}s - z\right) = \left(\frac{1}{2}s - a\right)^2 - \left(\frac{1}{2}s - a\right)x^2,$$

i.e.

$$y^2 + z^2 - \frac{2}{s}(s - a)yz - 2a(y + z) + a^2 = 0,$$

or reducing

$$(y + z - a)^2 = 4\left(1 - \frac{a}{s}\right)yz,$$

and we have thus three equations:

$$y + z + 2\sqrt{1 - \frac{a}{s}} \sqrt{yz} = a,$$

$$z + x + 2\sqrt{1 - \frac{b}{s}} \sqrt{zx} = b,$$

$$x + y + 2\sqrt{1 - \frac{c}{s}} \sqrt{xy} = c,$$

which are given by Schellbach in the same volume, p. 29; and it was for the sake of facilitating the comparison that the notation has been altered: in the case of the spherical triangle. To solve the system, Schellbach writes

$$a = s \sin^2 \phi, \quad b = s \sin^2 \chi, \quad c = s \sin^2 \psi,$$

reducing the equations to

$$y + z + 2\sqrt{yz} \cos \phi = s \sin^2 \phi,$$

$$z + x + 2\sqrt{zx} \cos \chi = s \sin^2 \chi,$$

$$x + y + 2\sqrt{xy} \cos \psi = s \sin^2 \psi,$$

whence, putting

$$\phi + \chi + \psi = 2\sigma,$$

the equations are satisfied by

$$x = s \sin^2(\sigma - \phi), \quad y = s \sin^2(\sigma - \chi), \quad z = s \sin^2(\sigma - \psi),$$

which leads to a simple geometrical construction. And if we substitute for ϕ, χ, ψ, σ , their values, it is easy to obtain

$$\begin{aligned} x &= \frac{1}{2} \left[1 - \sqrt{\left(1 - \frac{a}{s}\right)\left(1 - \frac{b}{s}\right)\left(1 - \frac{c}{s}\right)} \right] \\ &\quad + \sqrt{1 - \frac{b}{s}} \sqrt{\frac{a}{s}} \sqrt{\frac{b}{s}} \sqrt{1 - \frac{c}{s}} + \sqrt{1 - \frac{c}{s}} \sqrt{\frac{a}{s}} \sqrt{\frac{c}{s}} \sqrt{1 - \frac{b}{s}}, \\ y &= \frac{1}{2} \left[1 - \sqrt{\left(1 - \frac{a}{s}\right)\left(1 - \frac{b}{s}\right)\left(1 - \frac{c}{s}\right)} \right] - \sqrt{1 - \frac{a}{s}} \sqrt{1 - \frac{b}{s}} \sqrt{\frac{a}{s}} \sqrt{\frac{c}{s}} \sqrt{1 - \frac{c}{s}} \dots, \\ z &= \frac{1}{2} \left[1 - \sqrt{\left(1 - \frac{a}{s}\right)\left(1 - \frac{b}{s}\right)\left(1 - \frac{c}{s}\right)} \right] - \sqrt{1 - \frac{a}{s}} \sqrt{1 - \frac{b}{s}} \sqrt{1 - \frac{c}{s}} \dots, \end{aligned}$$

values which are also at once obtained from the formulæ in my paper above referred to. It may be remarked that the above equations for the determination of x, y, z (the distances of the points of contact from the adjacent angles of the triangle) are very similar in form to those given in the same paper for the determination of X, Y, Z , the radii of the inscribed circles.

171.

NOTE ON MR SALMON'S EQUATION OF THE ORTHOTOMIC CIRCLE.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 242–244.]

LET $U_1 = 0$, $U_2 = 0$, $U_3 = 0$ be the equations of three circles, and let V be the functional determinant of U_1 , U_2 , U_3 , the functions being in the first instance made homogeneous by the introduction of a variable z , which is ultimately replaced by unity; then the equation of the circle cutting at right angles the three given circles, or, as it may be called, the orthotomic circle, is $V = 0$. This elegant theorem of Mr Salmon's is connected with the theory developed by Hesse in the memoir, "Über die Wendepuncte der Curven dritter Ordnung," *Crelle*, t. XXVIII. (1844), p. 97).

In fact, let $U_1 = 0$, $U_2 = 0$, $U_3 = 0$ be the equations of three conics, the locus of a point such that its polars with respect to each of these conics, or indeed with respect to any conic having for its equation $\lambda U_1 + \mu U_2 + \nu U_3$ (where λ , μ , ν are arbitrary), pass through the same point, is a curve of the third order $V = 0$, where V is the functional determinant of U_1 , U_2 , U_3 .

Conversely, if the curve of the third order $V = 0$ be given, and U be a function of the third order, such that the functional determinant of $\frac{dU}{dx}$, $\frac{dU}{dy}$, $\frac{dU}{dz}$, or what is the same thing, the "Hessian" of the function U is equal to V , a condition which may be written $V = H(U)$, then we may take for the conics any three conics the equations of which are of the form $\lambda \frac{dU}{dx} + \mu \frac{dU}{dy} + \nu \frac{dU}{dz} = 0$. The equation $V = H(U)$ affords the means of determining U ; in fact, we shall have $U = aV + bH(V)$, where a and b are constants to be determined. This gives $H(U) = H(aV + bH(V)) = AV + BH(V)$, where A and B are given functions of a , b (a practical method of determining these functions was first given in Aronhold's memoir, "Zur Theorie der homogenen Functionen dritten Grades von zwei Variabeln," *Crelle*, t. XXXIX. (1850), pp. 140–159); and we have therefore

$V = AV + BH(V)$, i.e. $A = 1$, $B = 0$: the latter equation determines, what is alone important, the ratio $b : a$; the equation is of the third order, so that there are in general three distinct solutions $U = aV + BH(V) = 0$.

In the particular case in which the curves of the third order $V = 0$ is made up of a line $P = 0$ and a conic $W = 0$, i.e. where $V = PW = 0$, the curve $H(V) = 0$ is made up of the same line $P = 0$ and of a conic having double contact with the conic $W = 0$ at the point of intersection with the line $P = 0$, i.e. $H(PW) = P(lW + mP^2)$. And $U = aPW + bH(PW)$ is consequently a function of the same form, i.e. the cubic $U = 0$ is made up of the line $P = 0$ and of a conic having double contact with the conic $W = 0$ at its points of intersection with the line $P = 0$. We may therefore write $U = P(fW + gP^2)$, and forming with this value the equation $V = H(U)$ it may be noticed that, owing to the occurrence of a special factor which may be rejected, the resulting equation $G = 0$ gives only a single value for the ratio $f : g$. Forming from the value $U = P(fW + gP^2)$, the equation $\lambda \frac{dU}{dx} + \mu \frac{dU}{dy} + \nu \frac{dU}{dz} = 0$, the equation thus obtained will be of the form $W + 2PQ = 0$, which is the equation of a conic passing through the points of intersection of the line and conic $P = 0$, $W = 0$, and besides intersecting the conic $W = 0$ in two other points. And it may also be shown that the four points of intersection, (i.e. the points given by the equations $W = 0$, $W + 2PQ = 0$), the pole of the line $P = 0$ with respect to the conic $W = 0$, and the pole of this same line with respect to the conic $W + 2PQ = 0$, lie all six in the same conic. We see, therefore, that, given a curve of the third order, the aggregate of a line $P = 0$ and a conic $W = 0$, as the locus of the point such that its polars, with respect to three several conics (or a system depending on three conics), meet in a point, each conic of the system is a conic passing through the points of intersection of the line $P = 0$ and conic $W = 0$, and, moreover, such that the four points of intersection with the conic $W = 0$, and the poles of the line $P = 0$ with respect to the conic of the system, and with respect to the conic $W = 0$, lie all six in the same conic. In the particular case in which the line and conic $P = 0$, $W = 0$ are the line at ∞ and a circle, each conic of the system is a circle such that its points of intersection with the circle $W = 0$ and the centres of the two circles lie in a circle, i.e. the conics are circles cutting at right angles the circle $W = 0$, which agrees with Mr Salmon's theorem.

To verify the assumed theorems in the case of the curve of the third order

$V = PW = 0$, we may take

$$P = \alpha x + \beta y + \gamma z = 0$$

for the equation of the line, and

$$W = x^2 + y^2 + z^2 = 0$$

for the equation of the conic. I write, for greater convenience, $U = P(\frac{1}{2}fW + \frac{1}{2}gP^2)$; the Hessian of this is

$$\begin{vmatrix} f(P + 2\alpha x) + g\alpha^2 P, & f(\beta x + \alpha y) + g\alpha\beta P, & f(\alpha z + \gamma x) + g\gamma\alpha P \\ f(\beta x + \alpha y) + g\alpha\beta P, & f(P + 2\beta y) + g\beta^2 P, & f(\gamma y + \beta z) + g\beta\gamma P \\ f(\alpha z + \gamma x) + g\gamma\alpha P, & f(\gamma y + \beta z) + g\beta\gamma P, & f(P + 2\gamma z) + g\gamma^2 P \end{vmatrix},$$

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which is equal to

$$f^3 P(4P^2 - (\alpha^2 + \beta^2 + \gamma^2) W) + f^2 g P^3 (\alpha^2 + \beta^2 + \gamma^2) P^2,$$

i.e. we must have $4f + g(\alpha^2 + \beta^2 + \gamma^2) = 0$, or putting $g = -24$ and $\therefore f = 6(\alpha^2 + \beta^2 + \gamma^2)$, we have

$$U = P(3(\alpha^2 + \beta^2 + \gamma^2) W - 4P^2).$$

Forming the function $\lambda \frac{dU}{dx} + \mu \frac{dU}{dy} + \nu \frac{dU}{dz}$, and dividing by the constant factor $3(\alpha^2 + \beta^2 + \gamma^2)(\lambda\alpha + \mu\beta + \nu\gamma)$, we have for the equation of any one of the conics

$$W + 2 \left[\frac{2(\lambda x + \mu y + \nu z)}{\lambda\alpha + \mu\beta + \nu\gamma} - \frac{4P}{\alpha^2 + \beta^2 + \gamma^2} \right] P = 0,$$

which may be written under the form $W + 2PQ = 0$, where

$$Q = ax + by + cz = \frac{\lambda x + \mu y + \nu z}{\lambda\alpha + \mu\beta + \nu\gamma} - \frac{2(\alpha x + \beta y + \gamma z)}{\alpha^2 + \beta^2 + \gamma^2}.$$

We have therefore $a\alpha + b\beta + c\gamma = 1 - 2 = -1$, i.e. $a\alpha + b\beta + c\gamma + 1 = 0$. And there is no difficulty in showing that, given the two conics

$$x^2 + y^2 + z^2 = 0,$$

$$x^2 + y^2 + z^2 + 2(\alpha x + \beta y + \gamma z)(ax + by + cz) = 0,$$

the condition in order that the four points of intersection and the poles, with respect to each conic, of the line $\alpha x + \beta y + \gamma z = 0$, may lie in a conic is precisely this equation $a\alpha + b\beta + c\gamma + 1 = 0$.

172.

NOTE ON THE LOGIC OF CHARACTERISTICS.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 257–259.]

THE conditions in order that an equation of the sixth degree

$$(a, b, c, d, e, f, g) (x, y)^6 = 0$$

may have five of its roots equal are

$$A = ae - 4bd + 3c^2 = 0,$$

$$B = af - 3be + 2cd = 0,$$

$$C = bf - 4ce + 3d^2 = 0,$$

$$D = ag - 9ce + 8d^2 = 0,$$

$$E = bg - 3cf + 2de = 0,$$

$$F = cg - 4df + 3e^2 = 0,$$

equivalent of course to four relations between the coefficients: among the connections of these equations are

$$fA - eB - bF + cE = 0,$$

$$(3e^2 - 2df)A - 2deB - ecD + 2cdE + (3c^2 - 2bd)F = 0.$$

The system is one of the tenth order. To verify this, I write first

$$(A, B, C, F) = (A, B, C, F, cE) = (A, B, C, F, c) + (A, B, C, F, E),$$

i.e. the equations $A = 0, B = 0, C = 0, F = 0$ imply (by the first of the connectives) the additional equation $cE = 0$, viz. the system $A = 0, B = 0, C = 0, F = 0, cE = 0$, or what is the same thing, one of the systems $A = 0, B = 0, C = 0, F = 0, c = 0$, and $A = 0, B = 0, C = 0, F = 0, E = 0$.

We have in like manner

$$(A, B, C, F, E) = (A, B, C, F, E, ceD) = (A, B, C, F, E, D)$$

since (A, B, C, F, E, c) and (A, B, C, F, E, e) respectively vanish as being each of them equivalent not to four but to five relations, and therefore as not adding to the order of the system.

Again,

$$\begin{aligned} (A, B, C, c) &= (A, B, C, c, bF) \\ &= (A, B, C, c, b) + (A, B, C, c, F) \\ &= (ae, af, d^2, b, c) + (A, B, C, c, F). \end{aligned}$$

But here

$$\begin{aligned} (ae, af, d^2, b, c) &= (e, af, d^2, b, c) + d^2(a, af, d, b, c) \\ &= d^2(a, af, d, b, c), \end{aligned}$$

(for (e, af, d^2, b, c) vanishes as being equivalent to five relations, and therefore as not adding to the order of the system),

$$\begin{aligned} &= d(a, a, d^2, b, c) + d(a, f, d^2, b, c) \\ &= d(a, d^2, b, c), \end{aligned}$$

since, (a, f, d^2, b, c) vanishes for the above-mentioned reason.

Hence

$$(A, B, C, c) = d(a, d^2, b, c) + (A, B, C, c, F).$$

We have consequently

$$\begin{aligned} (A, B, C, D, E, F) &= (A, B, C, E, F) \\ &= (A, B, C, F) - (A, B, C, F, c) \\ &= (A, B, C, F) - \{(A, B, C, c) - (a, b, c, d^2)\}, \end{aligned}$$

which may be thus interpreted: the system (A, B, C, c) contains the system (a, b, c, d^2) , or what is the same thing, contains twice-over the system (a, b, c, d) . Discarding this contained system, the remainder of the system (A, B, C, c) is contained in the system (A, B, C, F) , and the residue of the

last-mentioned system is the system (A, B, C, D, E, F) , i.e. the system represented by the equations which express the equality of five roots of the given equation of the sixth degree.

It follows immediately that the order of the system (A, B, C, D, E, F) is $16 - (8 - 2) = 10$, i.e. that the system is, as above stated, one of the tenth order. The preceding is, I think, a good example of the kind of reasoning to be employed in what Mr Sylvester has most happily termed the Logic of Characteristics.

173.

ON LAPLACE'S METHOD FOR THE ATTRACTION OF ELLIPSOIDS.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 285–300.]

[THE method referred to is that given in Book III. of the *Mécanique Céleste*, Ed. I. 1798.] Let the equation of the surface of the ellipsoid be

$$lx^2 + my^2 + nz^2 - k = 0,$$

and take a, b, c as the coordinates of the attracted point, which is supposed to be exterior to the surface. Imagine an indefinitely thin cone, having the attracted point for vertex, and intersecting the surface of the ellipsoid; let ξ, η, ζ be the direction-cosines of the cone, and dS its spherical angle. Then if for a moment r denote the radius vector corresponding to a point within the ellipsoid, the element of the mass within the cone is $r^2 dr dS$; and we may thence, by an integration with respect to r , find the attractions parallel to the axes (and tending towards the centre) and the potential of the mass within the cone, viz. if r', r'' be the values of r at the surface of the ellipsoid, the attractions are

$$(r'' - r') \xi dS, \quad (r'' - r') \eta dS, \quad (r'' - r') \zeta dS,$$

and the potential is

$$\frac{1}{2} (r''^2 - r'^2) dS.$$

And putting for shortness

$$L = l\xi^2 + m\eta^2 + n\zeta^2,$$

$$I = la\xi + mb\eta + nc\zeta,$$

$$P = la^2 + mb^2 + nc^2 - k,$$

$$R = I^2 - PL,$$

then r', r'' will be the roots of the equation $Lr^2 - 2Ir + P = 0$, and we have consequently $r' = \frac{I - \sqrt{R}}{L}$, $r'' = \frac{I + \sqrt{R}}{L}$. We have, therefore, for the attractions parallel to the axes